OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 34 OF 55

AI-BASED ROAD DAMAGE DETECTION AND SEVERITY CLASSIFICATION USING YOLO AND DEEP LEARNING

Domain: Civil Infrastructure & Edge Computer Vision | **Module:** project_34_road_damage

OFFICIAL ABSTRACT

Road infrastructure maintenance is critical for ensuring public safety, reducing vehicle wear, and optimizing municipal budgets. Traditional road inspection methods rely on manual surveys or dedicated scanning vehicles, which are slow, labor-intensive, and expensive. To address these challenges, this study proposes an AI-based road damage detection and severity classification system utilizing the YOLO (You Only Look Once) framework combined with deep learning architectures. The system processes real-time road images or video feeds captured by standard vehicle-mounted cameras to identify various damage types, including longitudinal cracks, transverse cracks, alligator cracking, and potholes. By leveraging the real-time processing capabilities of YOLO, the proposed model detects anomalies and extracts critical bounding boxes representing distress locations. The detected damages are subsequently analyzed by a classification network that assesses their severity levels (e.g., low, medium, high) based on distress size, depth, and structural impact. The model is trained and validated on a comprehensive dataset of road damage images, achieving high precision and rapid inference rates. Experimental results demonstrate that the proposed system outperforms conventional computer vision techniques in both detection accuracy and processing speed. The implementation of this automated solution enables continuous monitoring of road networks, providing municipal authorities with actionable, cost-effective data for prioritized road maintenance, thereby enhancing overall transportation safety and prolonging infrastructure life.

📌 Project Summary & System Specifications

Project Title	AI-BASED ROAD DAMAGE DETECTION AND SEVERITY CLASSIFICATION USING YOLO AND DEEP LEARNING

Specialization / Category	Civil Infrastructure & Edge Computer Vision
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5034`
Local Implementation	`project_34_road_damage`